

APPENDIX 4.1 ADT LIST SPECIFICATION

ADT List

A list is a linear collection of entries of a type **T** which allows duplicate elements. An entry may be added at a specified position or at the end of the list. The list position starts from 1.

add(T newEntry)

Description : Adds **newEntry** to the end of the list.
Postcondition : **newEntry** has been added to the end of the list.

boolean add(Integer newPosition, T newEntry)

Description : Adds **newEntry** at position **newPosition** within the list. Position 1 indicates the first entry in the list.
Precondition : **newPosition** must be between 1 to total entries + 1.
Postcondition : **newEntry** has been added to the indicated position of the list.
Returns : **true** if **newEntry** was successfully added to the list; **false** otherwise.

T remove(Integer givenPosition)

Description : Removes the entry at position **givenPosition** within the list.
Precondition : **givenPosition** must be between 1 to total entries.
Postcondition : The entry at position **givenPosition** has been removed from the list.
Returns : The entry that was removed from the list.

clear()

Description : Removes all entries from the list.
Postcondition : The list is now empty.

boolean replace(Integer givenPosition, T newEntry)

Description : Replaces the entry at position **givenPosition** with **newEntry**.
Precondition : **newPosition** must be between 1 to total entries.
Postcondition : The entry at **givenPosition** has been replaced with **newEntry**.
Returns : **true** if the entry at **givenPosition** was successfully replaced in the list; **false** otherwise.

T getEntry(Integer givenPosition)

Description : Retrieves the entry at position **givenPosition** in the list.
Precondition : **newPosition** must be between 1 to total entries.
Postcondition : The list remains unchanged.
Returns : The entry at position **givenPosition**.

boolean contains(T anEntry)

Description : Determines whether the list contains **anEntry**.
Postcondition : The list remains unchanged.
Returns : **true** if **anEntry** is in the list, or false if not.

int getNumberOfEntries()

Description : Gets the number of entries currently in the list.
Postcondition : The list remains unchanged.
Returns : The number of entries currently in the list.

boolean isEmpty()

Description : Determines whether the list is empty.
Postcondition : The list remains unchanged.
Returns : **true** if the list is empty, or false if not.

boolean isFull()

Description : Determines whether the list is full.
Postcondition : The list remains unchanged.
Returns : **true** if the list is full, or false if not.